

# Lab Report

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## Abstract:

This project focuses on designing and manufacturing a hydraulic robotic arm and a machined component. The aim is to apply basic manufacturing concepts and develop practical engineering skills. In Part I, a robotic arm is designed and built using lightweight materials and processes such as laser cutting and laser working and simulated using CAD/CAM software (Inventor). The arm uses a hydraulic system (syringes and tubes) to create movement and perform tasks like picking and placing marshmallows. In Part II, the first part of the project was rod (machining), cutting a piece of iron following calculated and required dimensions for every part and using various types of machining. All the processes were done under the supervision of both module leader (Dr.Hesham Mohammed) and lab supervisor (Eng. Khaled). This part focuses on accuracy, proper tool use, and understanding manufacturing steps. The project also includes planning, material selection, and measurement evaluation. Overall, it helps improve understanding of manufacturing processes, design, and practical workshop skills.

| WBSU0912 |                        | Task Description                                                     |  | Week 5                          |  |  |  |  |  |  | Week 6                      |  |  |  |  |  |  |
|----------|------------------------|----------------------------------------------------------------------|--|---------------------------------|--|--|--|--|--|--|-----------------------------|--|--|--|--|--|--|
| I-1      |                        |                                                                      |  | Sun Mon Tue Wed Thu Fri Sat Sun |  |  |  |  |  |  | Mon Tue Wed Thu Fri Sat Sun |  |  |  |  |  |  |
| 1.1      | Project Management     | Planning and resource allocation                                     |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.1.1    |                        | Allocate resources to tasks (Rooftop: 200-300mm, SISO: <25cm)        |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.1.2    |                        | Create Gantt chart to manage the 30-hour project workload            |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.1.1    |                        | Finalize the Bill of Materials (BOM)                                 |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.2      | Design & Engineering   | CAD and technical documentation                                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.2.1    |                        | Looking at the families of hydraulic arms and how they function      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.2.2    |                        | Develop 3D models for arms (Humerus, Ulna) and base                  |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.2.3    |                        | Generate orthographic: medial, superior, ulnar, and base             |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.3      | Fabrication & Assembly | Production of parts                                                  |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.3.1    |                        | Cut cardboard, glue, grapple, and arm structure                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.3.1    |                        | Calculate real-time machining w/ 3D print 19 times                   |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.3.1    |                        | Install s-y, ligament, and from transmission lines                   |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.4      | Testing & Validation   | Performance evaluation against fabric                                |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.4.1    |                        | Verify each assembly: 10mm to 200mm-300mm range                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.4.2    |                        | E.X: 100mm to 150mm, 150mm to 200mm, 200mm to 250mm                  |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.4.3    |                        | Confirm assembly is self-supporting, no external aid                 |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.5      | Final Documentation    | Final report and mission report                                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.5.1    |                        | Overview and accuracy of arm, and W/2 validation (10%)               |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 1.5.2    |                        | Finalize report with photos and traceability notes                   |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2        |                        | Machine tooling & CAM                                                |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.1      | Machine tooling & CAM  | Machine tooling the component                                        |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.1.1    |                        | Grinding or CNC lathe                                                |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.1.2    |                        | Center lathe                                                         |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.1.1    |                        | Drilling                                                             |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.1.4    | Inventor Setup         | Milling                                                              |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.1.5    |                        | Threading                                                            |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.2      |                        | Inventor Setup                                                       |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.2.1    |                        | Inventor Setup                                                       |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.2.2    | Welding Tutorial 1 & 2 |                                                                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3      | CAM Processing         | CAM Processing                                                       |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.1    |                        | Setting up the workpiece                                             |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.2    |                        | Creating a flat end and tool: the end of the NO: 1 plate             |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.3    |                        | Creating a 10mm hole                                                 |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.4    | Inventor Setup         | Refining the diameter of the workpiece                               |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.5    |                        | Achieving the joint diameter and surface quality                     |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.6    |                        | Creating a groove in the rod                                         |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.7    |                        | Removing the glue from the surface from one end                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.8    | Inventor Setup         | Threading the hole in the surface from one end                       |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.9    |                        | Creating a 10mm hole                                                 |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.10   |                        | Creating a 10mm hole                                                 |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.11   |                        | Creating a 10mm hole                                                 |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.12   | Inventor Setup         | Tapping: M10 hole                                                    |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.13   |                        | Rotating the plate                                                   |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.14   |                        | Removing the extra 30mm stock                                        |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.3.15   |                        | Repeating the process on the other end of the workpiece (E.g. 2.3.3) |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.4      | Testing & Validation   | Simulation Test                                                      |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.4.1    |                        | Verifying the process                                                |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.4.2    |                        | Verifying the process                                                |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.4.3    |                        | Verifying the process                                                |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.5      | Final Documentation    | Final Documentation                                                  |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.5.1    |                        | Setup Sheet                                                          |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.5.2    |                        | Post-Process                                                         |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |
| 2.5.3    |                        | Creating the final report                                            |  |                                 |  |  |  |  |  |  |                             |  |  |  |  |  |  |

## Introduction for project 1:

In this part of the project, a hydraulic-powered robotic arm is designed and built to show the basic manufacturing and engineering principles. The aim is to create a working model that can move, lift and transfer 3 marshmallows using hydraulic pressure. The robotic arm uses syringes and tubes to control movement through fluid syringe. It is made from lightweight material like cardboard, using processes like laser cutting. These methods allow for easy design and construction of the different parts. During the project, important factors such as safety and correct use of tools and proper assembly are taken to produce a fully operated hydraulic robotic arm without having any serious injuries. The final robotic arm is expected to perform specific tasks including picking up and moving objects smoothly and accurately. This part helps develop practical skills in design, manufacturing, and teamwork, while also showing how hydraulic systems can be used in simple robotic operations.

## Limits:

- Max base length = 25 cm
- Reach Range ( arm reach ) = 200 to 300 mm

## BOM (Bill of material):

Bill of Material: Final Assembly of the hydraulic arm

| Part Number | Name            | Component Description   | Quantity | Type     |
|-------------|-----------------|-------------------------|----------|----------|
| 1           | Arm Section     | Assembly of the arm     | 1        | Assembly |
| 2           | Gripper Section | Assembly of the Gripper | 1        | Assembly |

Bill of Material: 1

| Part Number | Name                          | Component Description                                     | Quantity | Type     |
|-------------|-------------------------------|-----------------------------------------------------------|----------|----------|
| A           | 250 x 250 Base Support Bottom | Bottom base support                                       | 1        | Part     |
| B           | 250 x 250 Base Support        | Top base support                                          | 1        | Part     |
| C           | 80 x 80 mm Arm Base           | The base that will rotate around the pivot                | 1        | Part     |
| D           | 4 x 12 x 8 mm Arm Base        | Arm base                                                  | 2        | Part     |
| E           | 200 x 40 mm Arm               | Acts as the humerus of the arm                            | 2        | Part     |
| F           | Toothpick                     | This acts as a pivot for the arm segments to rotate about | 6        | Part     |
| G           | Toothpick 10 x 10 mm support  | Makes sure the toothpick does not fall                    | 4        | Part     |
| H           | 250 x 40 Arm                  | Acts as the radius or the ulna of the arm                 | 2        | Part     |
| I           | Syringe 10 ml                 | Syringe for the hydraulic function                        | 8        | Assembly |
| L           | Syringe holder                | Holds the syringe in place                                | 10       | Part     |
| M           | Syringe House                 | houses the syringe used for rotation                      | 2        | Part     |
| V           | Washer                        | ISO 7089 WASHER 8x16 STEEL GRADE A PLAIN NORMAL SERIES.1  | 1        | Part     |
| W           | Nut                           | ISO 4032 NUT M10 STEEL GRADE A HEXAGON.1                  | 1        | Part     |
| Y           | Bolt                          |                                                           | 1        | Part     |

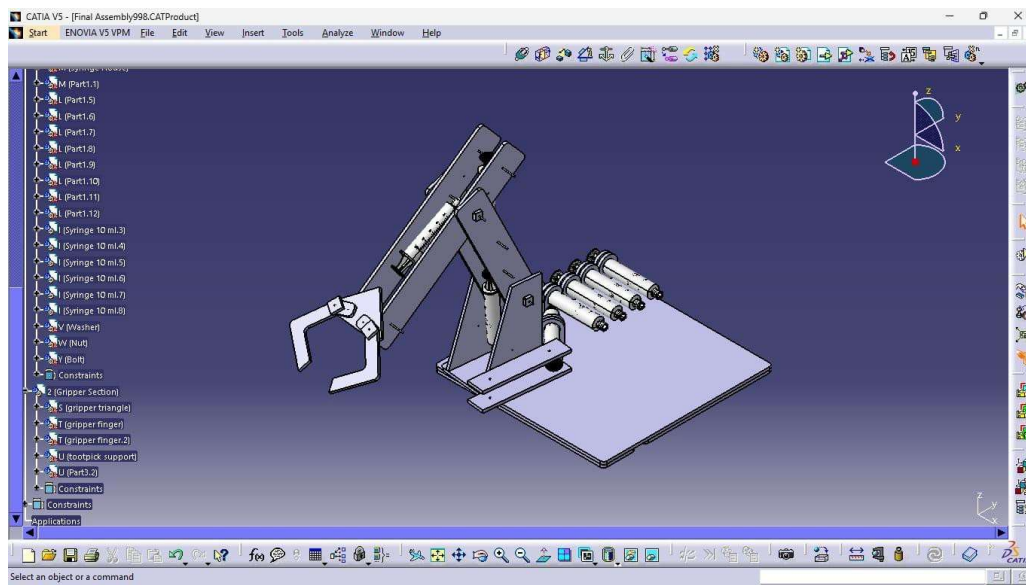
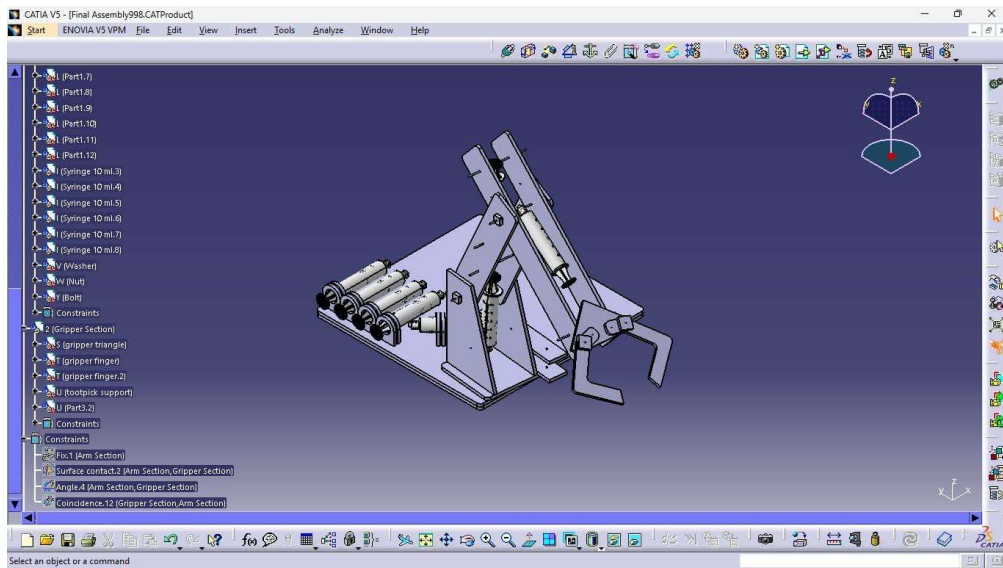
Bill of Material: I

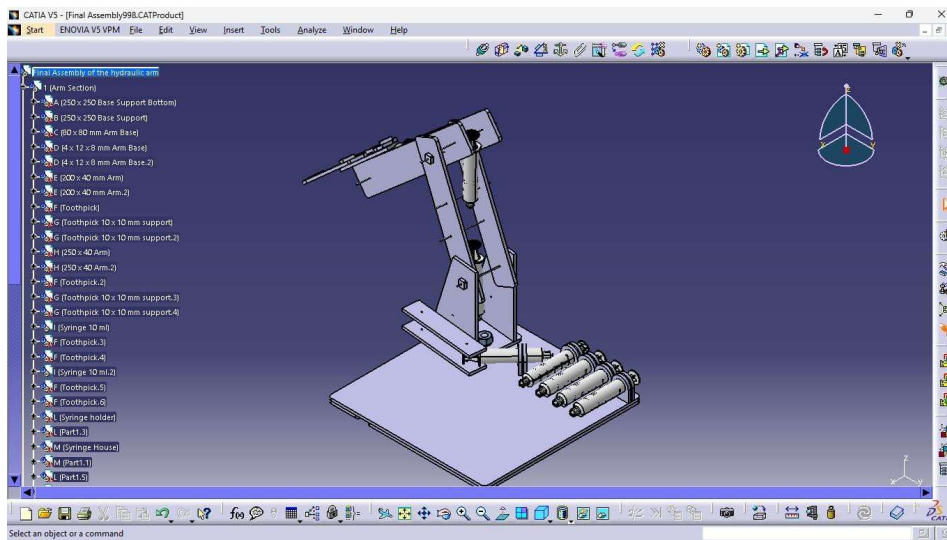
| Part Number | Name                  | Component Description | Quantity | Type |
|-------------|-----------------------|-----------------------|----------|------|
| N           | Rubber tampon 10ml    |                       | 1        | Part |
| P           | Syringe piston 10ml   |                       | 1        | Part |
| R           | Syringe cylinder 10ml |                       | 1        | Part |

Bill of Material: 2

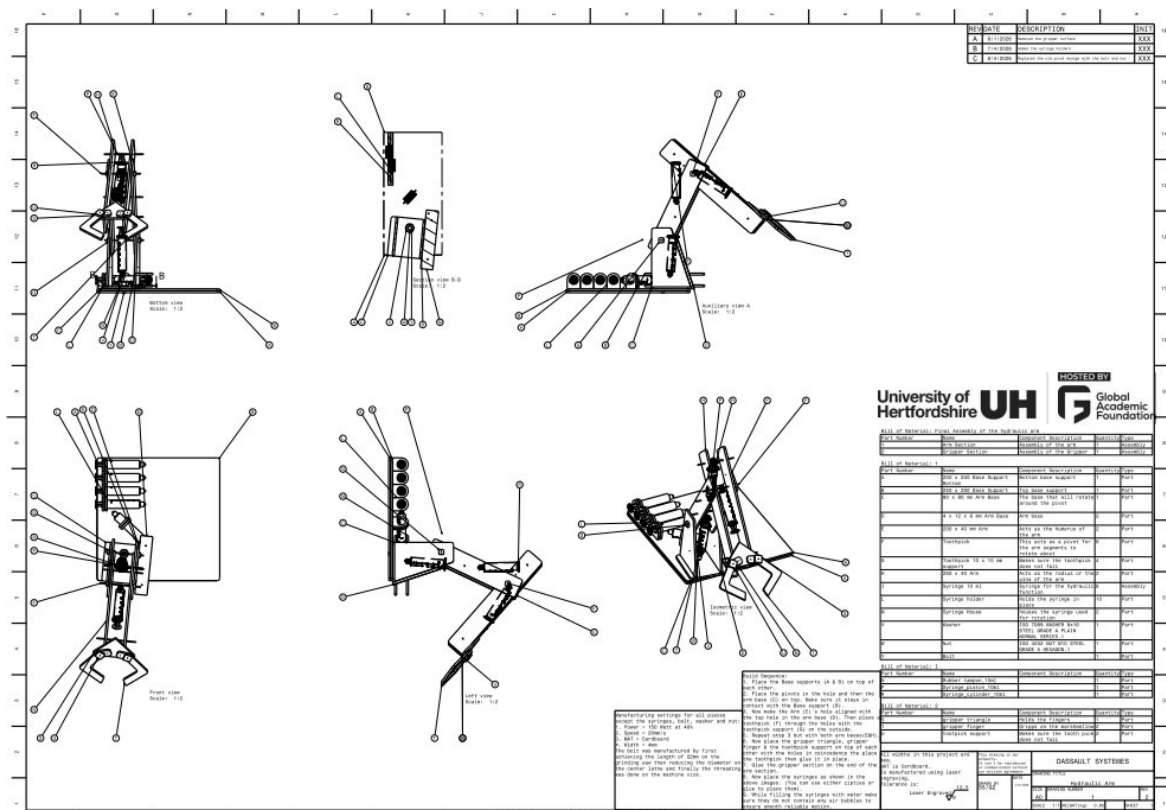
| Part Number | Name              | Component Description                   | Quantity | Type |
|-------------|-------------------|-----------------------------------------|----------|------|
| S           | gripper triangle  | Holds the fingers                       | 1        | Part |
| T           | gripper finger    | Gripes on the marshmallow               | 2        | Part |
| U           | toothpick support | Makes sure the tooth pick does not fall | 2        | Part |

## 3D model:





## Manufacturing and assembly procedures:



### **Build Sequence (Assembly procedures):**

1. Place the Base supports (A & B) on top of each other.
2. Place the pivots in the hole and then the arm base (C) on top. Make sure it stays in contact with the Base support (B).
3. Now make the Arm (E)'s hole aligned with the top hole in the arm base (D). Then place a toothpick (F) through the holes with the toothpick support (G) on the outside.

4. Repeat step 3 but with both arm bases(E&H).
5. Now place the gripper triangle, gripper finger & the toothpick support on top of each other with the holes in coincidence the place the toothpick then glue it in place.
6. Glue the gripper section on the end of the arm section.
7. Now place the syringes as shown in the above images. (You can use either zip ties or glue to place them).
8. While filling the syringes with water make sure they do not contain any air bubbles to ensure smooth reliable motion.

## Manufacturing Procedures:

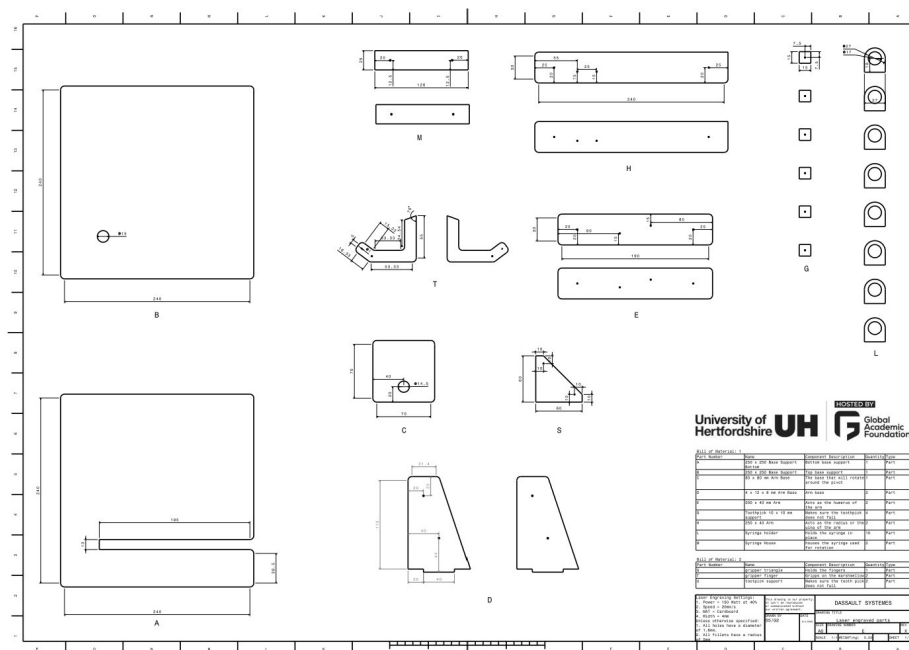
The bolt was manufactured by first achieving the length of 22mm on the grinding saw then reducing the diameter on the center lathe and finally the threading was done on the machine vice.

## Manufacturing Settings:

Manufacturing settings for all pieces except the syringes, bolt, washer and nut:

1. Power = 150 Watt at 40%
2. Speed = 20mm/s
3. MAT = Cardboard
4. Width = 4mm

## Laser engraving parts:

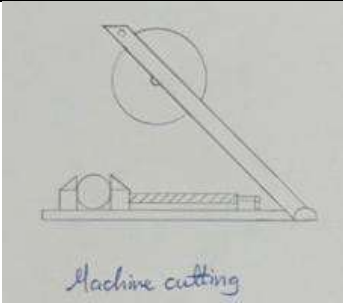
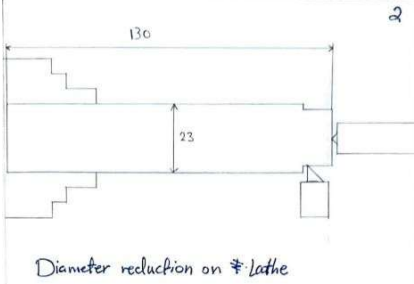
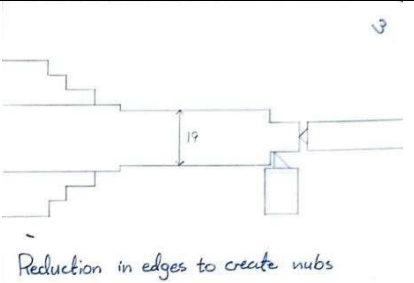


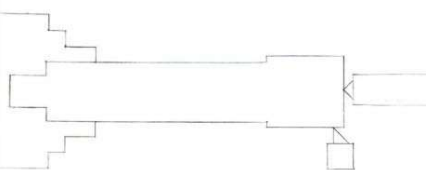
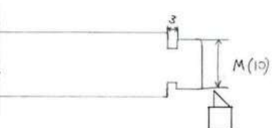
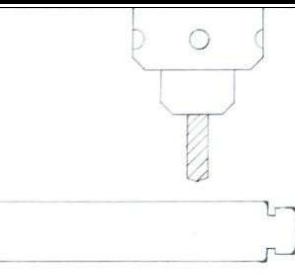
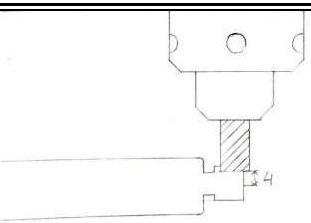
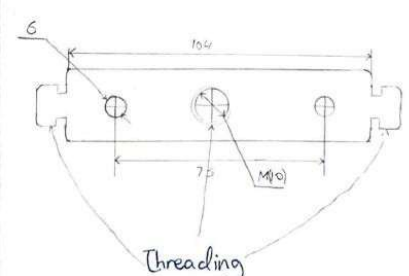


## Introduction for part 2:

This project is divided into 2 stages, Machining and CAM (Computer Aided Manufacturing). In first stage (Machining), we started by cutting the low carbon steel piece following certain and calculated dimensions. Followed by the CAM stage where the part is recreated on inventor CAM to be produced by CNC .

## Routing Sheet:

| <u>Op. No.</u> | <u>Sketch</u>                                                                                                                   | <u>Op. Description</u>                                                                                                       | <u>Resource (Machine tool)</u>         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| <u>1</u>       | <br><i>Machine cutting</i>                    | The work piece is placed under a sawmill for cutting process                                                                 | Saw machine & machine vice             |
| <u>2</u>       | <br><i>Diameter reduction on Lathe</i>       | The work piece is placed in a 3 jaw chuck for turning process using a cutting tool to reduce the diameter of the work piece. | Centre lathe, 3 jaw chuck & tail stock |
| <u>3</u>       | <br><i>Reduction in edges to create nubs</i> | The work piece gets stabilized by tailstock. The turning process is used to reduce diameters for the edges to create nubs    | Centre lathe, 3 jaw chuck & tail stock |

|                        |                                                                                                                                       |                                                                                |                                                                      |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <p><b><u>4</u></b></p> | <p>4</p>  <p>Repeat steps 2 and 3 for other half</p> | <p>Step 2 and 3 are repeated to create the other half</p>                      | <p>Centre lathe,<br/>3 jaw chuck &amp; tail stock</p>                |
| <p><b><u>5</u></b></p> | <p>5</p>  <p>Chamfering</p>                          | <p>Chamfering process is used to smooth the edges of the nubs</p>              | <p>Centre lathe,<br/>3 jaw chuck</p>                                 |
| <p><b><u>6</u></b></p> | <p>6</p>  <p>Drilling of 3 holes</p>                 | <p>Drilling process is used to create 3 holes on the top of the work piece</p> | <p>Driller &amp; Machine Vice</p>                                    |
| <p><b><u>7</u></b></p> | <p>7</p>  <p>Milling on one side of nubs</p>       | <p>Milling process is done by a cutting tool to flat the top of both nubs</p>  | <p>Cutting tool (Driller and Milling Machine) &amp; machine vice</p> |
| <p><b><u>8</u></b></p> | <p>8</p>  <p>Threading</p>                         | <p>Both Nubs and central hole got threaded by a tab wrench and die Stock</p>   | <p>Tab wrench, Die Stock &amp; machine vice</p>                      |





### **Time Evaluation formula for step:**

$$L/f \times N = \text{Time}$$

$L$  = total length = Length of workpiece + Approach length + Over-travel

Length of workpiece = 130mm

Approach length = 3mm

Over-travel length = 2mm

$N$  = 630rpm

$F$  = 0.15 mm/rev for roughing

$$135 / 0.15 \times 630 = 1.43 \text{ minutes}$$

### **Time evaluation comparison:**

In the CAM software, the total estimated time is 18 minutes and 27 second, the actual time spent in the workshop was 4 hours. Multiple reasons can explain this gap:

1. The CAM software assumes that the workpiece is set at the zero point, however, in reality time was spent making touches on the workpiece to achieve point zero.
2. We also frequently stopped machining to make sure the measurements were correct.
3. The CAM software doesn't include time spent on tool changes and movement of cutting tool into and away from workpiece.



4. Some time was also lost because the drilling machine was not operating properly.

5. Finally we were still learning therefore some time was spent learning how to operate the different machines.

## **CAM Procedures:**

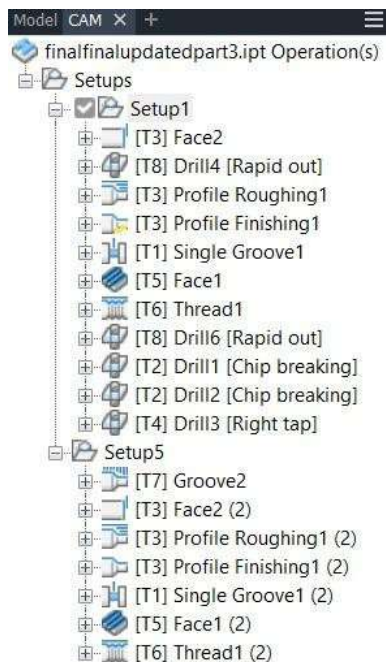
First the axis and stock were setup with 30 mm extra stock at the back to be held by 3 jaw chuck. The stock was set up with same dimensions as the machining stock (23 mm diameter, 161 mm length).

- Turning face: To touch the end part of the workpiece and create a flat end.
- Drilling 1 (Centring): Creating a centre at the end of the workpiece for tailstock.
- Profile roughing and finishing: To achieve diameter and surface quality.
- Single grooving: Creating the 3 mm groove.
- Face milling: Removing 1 mm of material at the horizontal surface.
- Threading: To thread the end of workpiece.
- Drilling 2 (Centring): Creating centre to drill the holes.
- Drilling 3 & 4: To create M10 and 6 mm diameter holes.
- Tapping: To tap the M10 holes.

Another setup is done and workpiece is rotated.

- Grooving: Removing extra 30 mm stock.
- Repeating processes for the other end of the rod.

Finally simulation is done and setup sheet and G code are produced.



## **Setup sheet:**

```

Joa 0E:11C1P110H: SetUp1
O 0000000000000000
C:\Users\Asus\Desktop\final\finalupdatedpart3. ipi

```

WCS#0  
 S1ac11;  
 OX: 231m1  
 DY 23mm  
 DZ 161mm  
 p,mc  
 DX: 1'9mm  
 DY 19mm  
 DZ: 130mm  
 STOCK LOWER IN WCS.α  
 X: -11.5mm  
 Y -11.5mm  
 Z:-161mm  
 STQ:1X UF-IER. INWCS #0  
 X: 11.5mm  
 y 11.5Yrtf  
 l 0mm

NUMBER OF OPERATIONS: 18  
 NUMBER OF TURNS: 8  
 TOOL: T1 T2 T3 T4 T5 T6 T7 T8  
 MATERIAL FEED: 1600 mm/min  
 MAXIMUM FORCE: 5000 N  
 CLIPPING: 1, NC# 1212 A9mm  
 APPROXIMATE: 2797.0 mm  
 ESTIMATE: 18m:27r

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 11 01   | <p> <i>N</i>[Fe3]OO<sup>+</sup>lurim<sup>+</sup>l<br/>           INSERT &amp;_lare<br/>           WmThk:3mm<br/>           @D4      :11ON:1T:pigeofol<br/>           TOOL: ORENTJUCIN in Tube1: Racial<br/>           I.A.O.IIMNFEED 56.8 @ 1mm/min<br/>           m.o.IIMMSPINKE SPEED: 1600rpm<br/>           O.Jin:0.5r.,Nel.:30mm<br/>           fl.pm o1,t,tst: 234mm<br/>           ESTIMATECLOCK TIME: 1m42s (~2%)         </p> | <p>           HCl test:EximolalR9L1         </p> |
| 12 D2L2 | <p>           T<sup>+</sup>U<sup>+</sup>U<sup>+</sup>U<sup>+</sup>U<sup>+</sup><br/>           L1:11111tr: 5.57mm<br/>           TFRANGE: 118°<br/>           LENGTH 110.51ml<br/>           FwrElI:3<br/>           M.o.IIMNFEED 115mm/min<br/>           @-SPIL QASPL19: 12.77rpm<br/>           m.o.MOGISTANCE 104.30mm<br/>           RAPIDstvs.la: 204.2fmi<br/>           ESTIMATECLOCK TIME: 50g 14.87s         </p>            |                                                  |

|                          |                              |                     |
|--------------------------|------------------------------|---------------------|
| OpenE[0]n1/S.            |                              | T300                |
| Occid;RiRn...;T FKE12    | \$v;vAQIS<cdJ;Jf0m0m         | TVP 118r18f33 tminr |
| STRATEGY: TmTing Fact-1  | FEDRATE PER REV.0.1mm        | INSERT: ISO C 80d;g |
| WCS.#0                   | CLTfTmaDIT.11.1NC 33.69mm    | EDGE LENGTH: 4.8mm  |
| TcJf0C.0.C.mmm           | RiPcIDic;ANc.4.0.1           | Idc;AACbLc.0.8 mm   |
| M...L;UM :TEPOUVER;U.5mm | E:TWIJD CYcIcT.11>E JUS (3~) | CRISSE.Cf0t0tT      |
|                          | Coou.N: Flood                | TOLERANCE           |

|                     |                                 |                          |
|---------------------|---------------------------------|--------------------------|
| OpenLon2(18:        |                                 | T>DBLI                   |
| DIST:=PUSHING       | CUT:=SPH+DLE \$p.m.: 1600rpm    | TYPE: conical drill LIVE |
| SURATEGY: Drilling  | MAXIMUM FEEDRATE: 160mm/min     | WAGER: 10mm              |
| VCS #ID             | CUTTING FLUTE: 8mm              | LINFLUTE 11.5            |
| To.1(RANC): O.G.hom | R.u. TO CRYSTAL: 20mm           | TIP:nc; 50mm             |
|                     | ESTIMATED CRACK TIME: JS 10.311 | FLUTES J                 |
|                     | CoolAnt: Flood                  |                          |

|                                       |                             |
|---------------------------------------|-----------------------------|
| Operon:Sh3/1/S.                       | <b>T300</b>                 |
| OS:CC...P110...Infro66E...o...lghing1 | Typ...genC01...1.mn19       |
| WC5:80                                | INdE/AR.ISO.C 804050        |
| MLER.LINC.0.0.0mm                     | INdE/AR.LINC.4mm            |
| M...JMUM.E5.0.0mm                     | INdE/AR.CMB.0.0mm           |
| M...JMUM.E5.0.0mm                     | C...S...n.Y100...T          |
| M...JMUM.E5.0.0mm                     | TOLERANCE                   |
| M...JMUM.E5.0.0mm                     | RELIEF.34deg                |
| M...JMUM.E5.0.0mm                     | OS...110M.T10N.T0...tangent |
| M...JMUM.E5.0.0mm                     | DISCUPTION-CN-IT.Rish.H     |

|                                |                                    |                      |
|--------------------------------|------------------------------------|----------------------|
| Oper:JL3(411)                  | S.....c= SF Ew 35m/min             | TJJO                 |
| #E&1uF1uP1uN1uPror1uFinishing1 |                                    | TypE:90nc:all.1fring |
| WC5 01                         | FEEDRATE PER RE: 0.07mm            | INBER: ISO C OQdeg   |
| TOLER:ANCE:0.07mm              | CUTTING OBTANCE: 12E:27mm          | Ed&E:IKH-T4mm        |
| 1YN-1MUM1uP1uPC1u1.2m1u        | RnL:01511111149.21mm               | 1QM.....1..05.30m    |
|                                | T:V-TED CYC.ETH:14m:2m:541u(15.7%) | CowS'S SELECTED: T   |
|                                | COOLTR: Flood                      | TERMINATE: M         |

|                              |  |                                        |  |
|------------------------------|--|----------------------------------------|--|
|                              |  | RELIEF3dsig                            |  |
|                              |  | On-1: $\mu$ M -10M; Td: tang ent       |  |
|                              |  | DESCRIPTION: CN-Ft Right Hand          |  |
| OPERATIONS: 10               |  | T1 di                                  |  |
| DISCRIPTION: Sing Groove 1   |  | TYPE: groove 1a                        |  |
| C11: 11.0; T: 1.7; Tang Croc |  | 1.1; 1.1; 1.1; 1.1                     |  |
| WCS: 10                      |  | WOT11: 1mm                             |  |
| TOL: ERANCE: 0.01mm          |  | COMPENBATION: 7ip tangent              |  |
| STOOR: T1: 0.0; Item         |  | E: 11.1; UED: CVC: 11.1; 1.1; 1.1; 1.1 |  |
|                              |  | 89% - 100%                             |  |

|                             |                                   |                      |
|-----------------------------|-----------------------------------|----------------------|
| OperaJQn:16                 | MA:1:1:1:SPINDLE SPEED: 7000R/M   | TS DS LS             |
| OESCPIPIH:1F3ce1            | PAW:1:1:1:FEEDRATE: 120mm/min     | TYJJE [ij] eMm LLIV: |
| STRATJ3:1 FawWj             | CUTTIPIQ 0B1:11.NC30.15mm         | DWAETER.16mm         |
| WCS.#0                      | Rn-10 015*1.....CJ: 22.1mm        | ENHIE:B0mm           |
| To,1:1: Cj:1)mm             | E5:1F.V:1:0 CQJ:1:1:1:1214 (1.9%) | FWTU:1:              |
| MVik-UM STEROTJLJ:15-2YU1T1 | COLA1T: Flood                     |                      |

|       |                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                       |                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| T3 DO | <p>1-PE gonoral turn'd in</p> <p>ISO: ISO C 30deg</p> <p>EOH: LENQTH 4mm</p> <p>NOSE I (F1) (m): 0.8m/Y</p> <p>CROSSSECT:ort</p> <p>Tox.Er.HCEM</p> <p>RC:ICJ^A 30deg</p> <p>C&gt;U&gt;EASION: Txi langont</p> <p>DESCRIPTION CNMT Right Hand</p> | <p>TOOL ORIENTATION IN Txi M1: R3C1</p> <p>MAXXWAM FEED: 176.903mm/min</p> <p>IAr.XbXm SPk.10LE Speed: 1600rpm</p> <p>Cvrt1 M1: 0.1 T1 LJC: 62.12m/M</p> <p>RAPID DISTOLICE: 1054.38mm</p> <p>EBTWI_TED CYCLE TIME: 9m:45S (-311)</p> | <p>Hot.DER.ISO L R 30</p> |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|

T4 D-4L4  
TYPIE: right hand tap LIVE  
MA(.....)PINOUT: SIpTet 25Sp'n  
LENGTH: 50mm  
CURING DISTANCE: 18mm  
PLUTED: 3  
RAPID ON+TOLE: 14mm  
ESTIMATED CURE TIME: 21 sec

**TH D5L5**  
 TYPE: flat end mill LIVE  
 O.D.  ID: 16mm  
 LENGTH: 80mm  
 FUJTEB: 3  
 M...x- FEED: 12Emm/min  
 I.A.r.: <WWM-sPJNOUI S>@H: 70Clptn  
 CUNING DISTANCE: 60.88mm  
 RAPID DRILL wd: 43.4mm  
 E.S.The Jet, eve<OTC: 42, 0.8tj

|                            |                                       |                         |
|----------------------------|---------------------------------------|-------------------------|
| 16D6                       |                                       |                         |
| TYPE: thread turning       | TOOL ORIENT: JICIN 11° TURRET: Ra-Gal | Hot.Offt: Stra 1 g/hf 1 |
| ILIS: Jrt: ISO double      | M: JICRIUM Fe 8, Ometrid              |                         |
| COMID: JATION: Tq: langont | A: J-J: JUAL SPINOLE SpEm 5000rpm     |                         |
|                            | CUTTING DISTANCE: 239.43mm            |                         |
|                            | RAPID Dr: "mc: 800 U/min              |                         |
|                            | ESTIHAJ: JEOCY: LE Tr: U: 108 (0.11%) |                         |

|                           |  |                                      |               |
|---------------------------|--|--------------------------------------|---------------|
| T7 D7                     |  |                                      |               |
| TYPE: groove turner       |  | TOOL: C418HTA10N 1111TURRET: 13      | Head External |
| 11 S(*1): □□re            |  | IA: <0.018 FkC: 15j/min              |               |
| Wton L: 3mm               |  | PAWMIN SPINDLE SPEED: 160Chjm        |               |
| NOSE RADIUS: 0.2mm        |  | CUTTING DISTANCE: 142.52mm           |               |
| COMIENSAT10t: Tip tangen1 |  | RAPID DE: 1.0M/Cjt: 252.58mm         |               |
|                           |  | ES111AJ1: 1E00YLeTtoE: 1111 US (S,*) |               |

|                      |                                    |
|----------------------|------------------------------------|
| HI 00L6              | IAIXIAN FEED: 1600mm               |
| TYPE:centrdrill LIVE | Mol.: SPINIDUR3181D.: 1600pm       |
| DRILL: 10mm          | Ort.m: 0.51_mmc3 20mm              |
| TIPAmq; 1180         | RAPD 015r.01GE: 10mm               |
| LENGTH: 50mm         | ESTIV_TED CYCLE TIME: 41 S B.711/L |
| RWTE: 3              |                                    |

Operat.Crt111f.

DEL:RUPTION: TTM/cadit

STRATEGITiming Thread

WCS:#0

TCL:11□: 0.01mm

□:MMHSPIN:LESPEEO:400°pm

FEEDUITE PER REV:1 mm

Cum,io DISTANCE: 133.71mm

Wo-ID DI:TW: 352.14mm

EstMATED CYCLE TIME 4s. 10.4a.1

Ciclot.Abr. Flood

T60G

Type: m/Lad.t.t.□

INE:ER: ISO □

Q:IMPEN:TiCm: Tang.tangent

|                     |                              |                         |
|---------------------|------------------------------|-------------------------|
| Operatkit\$J.S.     | MA:10:11WSPIDE SPEED 000.00  | 1110VIL\$               |
| (RESCRIP11UN") 1196 | MAxV:1JMM FEEDRATE130mm/min  | TYPE etf10I" drill LIVE |
| SIRATEQ1Drilling    | CLJ1T0 DISTANCE: 18mm        | DIAMETER: 10mm          |
| WCS #0              | fw:10D1sr_Not:115mm          | L1P6.NAL:118.0          |
| TOUR1RAJ*: 0.01mm   | E*TIMATED CYCETIME 37s (3PL) | TP16M 50mm              |
|                     |                              | FLUES: J                |

Operator: 918,  
DESCAIP11UN.D11  
M...J...  
WCS.#0  
TouR/Mp1: 0.01mm  
M...J...  
M...J...  
TIP TPO DISTANCE: 35.18mm  
D 0151...NCL 46.5um  
ISUMATED CYCLETIME 16S (1.5%)  
Cilost.Astr. Flood  
T2@21.2  
TYP: cril Liv:  
DWAE1088.5mm  
TIPANILE: 116.5  
LININT: 11Q.5(nm  
FLUTE:S-J

|                   |                                    |                  |
|-------------------|------------------------------------|------------------|
| Operation: 10'18  |                                    | T2 D2L2          |
| OAS:C11PBN:Orat2  | MaxTt1tu11t1IND:eSIIUJ: t275rm     | TW:e; drill LIVE |
| SIRATQV: Drilling | MA:t11;ur,t11:EEDRAE: t27.5ml/vr,u | DIMME:Er6tm      |
| WC:5-N-0          | CUTTING:D191)1JoICE: 68.8mm        | TIP-KiaLE:118'   |
| T-RANCE:0.01mm    | RAPID DIH...N: 157.36mm            | (LEN)TH: 110.Smm |
|                   | =>BM'D:D CtuT11cm: 348 (J.T11)     | Fw-rn: J         |
|                   | CoA:ANT: FLOOJ:                    |                  |

Operation 111UL  
GDS:CHPION:1x3  
SIRAm:3: Drilling  
TVCS:40  
TOLERANCE:0.01mm

MADIWIU:INCILE SPIRE: 255,pm  
M...:1-JWJFEEDRA11: 381,92mm/min  
CLTJNG DIST:RICE 457m  
RSSD DRILLUC : 4mm  
STTMM-DCT:UT11:1m 6S (GJS)  
Coa:Aw: FLOOL

T: D4 L4  
TIME: right h3fnd sp LUE  
Dw:ETHEL 1CMm  
LENJOC 50mm  
FWTLE113

Operation 12118  
OleCfItts>RON Groove2  
SIRATE>Gv:Twinty Ptor.18: Groove rEE.BRTE PER AEv>0.08mm  
WC5.80  
TOLE.PANCE>0.01mm  
StIk1rol.w :□ 1mm

SurfT...all S.C.ID: Q5mmIn  
GUTTINGDIST.RICE: 142.92mm  
RAPID DWH.LICE: 282.3mm  
55.11%,TIP CnM1T1c=mm: 1mC5(□)1u  
Cn111111NS,11011:Tip target

T7D7  
TW11:1111O'e turning  
INSERT>Gv>111111  
WID1113mm  
ID1111R.Dhg,0\_2mm

|                                                                                                                                                                                 |                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                          |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Operation 13/18</b><br><b>DESCRIPTION:</b> Face2 (2)<br><b>STRATEGY:</b> Turning Face<br><b>WCS:</b> #0<br><b>TOLERANCE:</b> 0.01mm<br><b>MAXIMUM STEPOVER:</b> 0.5mm        | <b>SURFACE SPEED:</b> 30m/min<br><b>FEEDRATE PER REV:</b> 0.1mm<br><b>CUTTING DISTANCE:</b> 33.69mm<br><b>RAPID DISTANCE:</b> 159.66mm<br><b>ESTIMATED CYCLE TIME:</b> 37s (3.4%)<br><b>COOLANT:</b> Flood          | <b>T3 D0</b><br><b>TYPE:</b> general turning<br><b>INSERT:</b> ISO C 80deg<br><b>EDGE LENGTH:</b> 4mm<br><b>NOSE RADIUS:</b> 0.8mm<br><b>CROSS SECTION:</b> T<br><b>TOLERANCE:</b> M<br><b>RELIEF:</b> A 3deg<br><b>COMPENSATION:</b> Tip tangent<br><b>DESCRIPTION:</b> CNMT Right Hand |  |
| <b>Operation 14/18</b><br><b>DESCRIPTION:</b> Profile Roughing1 (2)<br><b>WCS:</b> #0<br><b>TOLERANCE:</b> 0.01mm<br><b>MAXIMUM STEPOVER:</b> 1.2mm                             | <b>SURFACE SPEED:</b> 30m/min<br><b>FEEDRATE PER REV:</b> 0.15mm<br><b>CUTTING DISTANCE:</b> 65.47mm<br><b>RAPID DISTANCE:</b> 216.28mm<br><b>ESTIMATED CYCLE TIME:</b> 43s (3.8%)<br><b>COOLANT:</b> Flood         | <b>T3 D0</b><br><b>TYPE:</b> general turning<br><b>INSERT:</b> ISO C 80deg<br><b>EDGE LENGTH:</b> 4mm<br><b>NOSE RADIUS:</b> 0.8mm<br><b>CROSS SECTION:</b> T<br><b>TOLERANCE:</b> M<br><b>RELIEF:</b> A 3deg<br><b>COMPENSATION:</b> Tip tangent<br><b>DESCRIPTION:</b> CNMT Right Hand |  |
| <b>Operation 15/18</b><br><b>DESCRIPTION:</b> Profile Finishing1 (2)<br><b>WCS:</b> #0<br><b>TOLERANCE:</b> 0.01mm<br><b>MAXIMUM STEPOVER:</b> 1.2mm                            | <b>SURFACE SPEED:</b> 35m/min<br><b>FEEDRATE PER REV:</b> 0.07mm<br><b>CUTTING DISTANCE:</b> 22.84mm<br><b>RAPID DISTANCE:</b> 163.88mm<br><b>ESTIMATED CYCLE TIME:</b> 23s (2.1%)<br><b>COOLANT:</b> Flood         | <b>T3 D0</b><br><b>TYPE:</b> general turning<br><b>INSERT:</b> ISO C 80deg<br><b>EDGE LENGTH:</b> 4mm<br><b>NOSE RADIUS:</b> 0.8mm<br><b>CROSS SECTION:</b> T<br><b>TOLERANCE:</b> M<br><b>RELIEF:</b> A 3deg<br><b>COMPENSATION:</b> Tip tangent<br><b>DESCRIPTION:</b> CNMT Right Hand |  |
| <b>Operation 16/18</b><br><b>DESCRIPTION:</b> Single Groove1 (2)<br><b>STRATEGY:</b> Turning Groove<br><b>WCS:</b> #0<br><b>TOLERANCE:</b> 0.01mm<br><b>STOCK TO LEAVE:</b> 0mm | <b>SURFACE SPEED:</b> 25m/min<br><b>FEEDRATE PER REV:</b> 0.05mm<br><b>CUTTING DISTANCE:</b> 15mm<br><b>RAPID DISTANCE:</b> 171mm<br><b>ESTIMATED CYCLE TIME:</b> 52s (4.7%)<br><b>COOLANT:</b> Flood               | <b>T1 D1</b><br><b>TYPE:</b> groove turning<br><b>INSERT:</b> Square<br><b>WIDTH:</b> 3mm<br><b>COMPENSATION:</b> Tip tangent                                                                                                                                                            |  |
| <b>Operation 17/18</b><br><b>DESCRIPTION:</b> Face1 (2)<br><b>STRATEGY:</b> Facing<br><b>WCS:</b> #0<br><b>TOLERANCE:</b> 0.01mm<br><b>MAXIMUM STEPOVER:</b> 15.2mm             | <b>MAXIMUM SPINDLE SPEED:</b> 700rpm<br><b>MAXIMUM FEEDRATE:</b> 126mm/min<br><b>CUTTING DISTANCE:</b> 30.73mm<br><b>RAPID DISTANCE:</b> 20.9mm<br><b>ESTIMATED CYCLE TIME:</b> 21s (1.9%)<br><b>COOLANT:</b> Flood | <b>T5 D5 L5</b><br><b>TYPE:</b> Ball end mill LIVE<br><b>DIAMETER:</b> 16mm<br><b>LENGTH:</b> 80mm<br><b>FLUTES:</b> 3                                                                                                                                                                   |  |
| <b>Operation 18/18</b><br><b>DESCRIPTION:</b> Thread1 (2)<br><b>STRATEGY:</b> Turning Thread<br><b>WCS:</b> #0<br><b>TOLERANCE:</b> 0.01mm                                      | <b>MAXIMUM SPINDLE SPEED:</b> 400rpm<br><b>FEEDRATE PER REV:</b> 1mm<br><b>CUTTING DISTANCE:</b> 105.71mm<br><b>RAPID DISTANCE:</b> 448.14mm<br><b>ESTIMATED CYCLE TIME:</b> 5s (0.5%)<br><b>COOLANT:</b> Flood     | <b>T6 D6</b><br><b>TYPE:</b> thread turning<br><b>INSERT:</b> ISO double<br><b>COMPENSATION:</b> Tip tangent                                                                                                                                                                             |  |

**G code:**

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G11 G12 G13 G14

G15 G16 G17 G18

G19 G20 G21 G22

G23 G24 G25 G26

G27 G28 G29 G30

G31 G32 G33 G34

G35 G36 G37 G38

G39 G40 G41 G42

G43 G44 G45 G46

G47 G48 G49 G50

G51 G52 G53 G54

G55 G56 G57 G58

G59 G60 G61 G62

G63 G64 G65 G66

G67 G68 G69 G70

G71 G72 G73 G74

G75 G76 G77 G78

G79 G80 G81 G82

G83 G84 G85 G86

G87 G88 G89 G90

G91 G92 G93 G94

G95 G96 G97 G98

G99 G100 G101 G102

G103 G104 G105 G106

G107 G108 G109 G110

G111 G112 G113 G114

G115 G116 G117 G118

G119 G120 G121 G122

G123 G124 G125 G126

G127 G128 G129 G130

G131 G132 G133 G134

G135 G136 G137 G138

G139 G140 G141 G142

G143 G144 G145 G146

G147 G148 G149 G150

G151 G152 G153 G154

G155 G156 G157 G158

G159 G160 G161 G162

G163 G164 G165 G166

G167 G168 G169 G170

G171 G172 G173 G174

G175 G176 G177 G178

G179 G180 G181 G182

G183 G184 G185 G186

G187 G188 G189 G190

G191 G192 G193 G194

G195 G196 G197 G198

G199 G200 G201 G202

CHRP (SPINULE ANTENNA)  
G1 G2 G3 G4 (SPINULE ANTENNA)  
G5 G6 G7  
G8 G9  
G10

G11 G12 G13 G14

G15 G16 G17 G18

G19 G20 G21 G22

G23 G24 G25 G26

G27 G28 G29 G30

G31 G32 G33 G34

G35 G36 G37 G38

G39 G40 G41 G42

G43 G44 G45 G46

G47 G48 G49 G50

G51 G52 G53 G54

G55 G56 G57 G58

G59 G60 G61 G62

G63 G64 G65 G66

G67 G68 G69 G70

G71 G72 G73 G74

G75 G76 G77 G78

G79 G80 G81 G82

G83 G84 G85 G86

G87 G88 G89 G90

G91 G92 G93 G94

G95 G96 G97 G98

G99 G100 G101 G102

G103 G104 G105 G106

G107 G108 G109 G110

G111 G112 G113 G114

G115 G116 G117 G118

G119 G120 G121 G122

G123 G124 G125 G126

G127 G128 G129 G130

G131 G132 G133 G134

G135 G136 G137 G138

G139 G140 G141 G142

G143 G144 G145 G146

G147 G148 G149 G150

G151 G152 G153 G154

G155 G156 G157 G158

G159 G160 G161 G162

G163 G164 G165 G166

G167 G168 G169 G170

G171 G172 G173 G174

G175 G176 G177 G178

G179 G180 G181 G182

G183 G184 G185 G186

G187 G188 G189 G190

G191 G192 G193 G194

G195 G196 G197 G198

G199 G200 G201 G202





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G1 X11.2 F04.

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## **Conclusion:**

In this assignment we increased our catia designing skills, learned how to laser engraved cardboard pieces and it's manufacturing settings. We achieved all success criteria. We learned machining operations and how to operate them using traditional machining and CAM software for CNC machines.

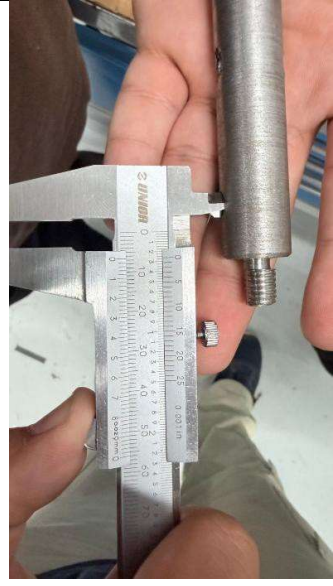
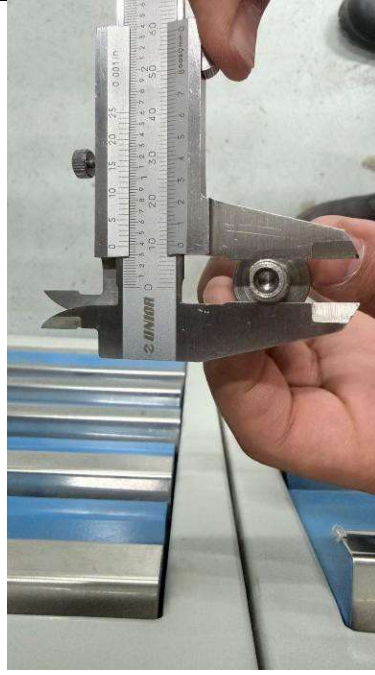
## **Reflective essay:**

In the future I would like to design the parts in a truss shape to reduce the material volume used and increase the smoothness of the hydraulic arm.

## **Appendices:**

Rod measurements:









### Extra photos for the rod machining:

